The following explanations are generated using the **ChatGPT (GPT-5.1 model)**.

1. The formula F (red_room & F (landmark_3)) expresses that eventually a state will occur where the agent reaches the red room, and from that point onward, it is guaranteed that landmark three will be visited at some later time. The outer F (Finally) operator indicates that this entire sequence, reaching the red room and then eventually reaching landmark three must occur at some time in the future. The conjunction (red_room & F(landmark_3)) specifies that when the red room is reached, there must still be a future moment in which the condition landmark_3 becomes true. Thus, the formula formally captures the directive in the natural language description "go to the red room, then go to landmark three" by requiring that the agent first achieves the state of being in the red room and subsequently reaches landmark three afterward.
2. The formula F (third_floor & F (green_room)) specifies that at some point in the future, the condition third floor will become true, and from that moment onward, there will exist a later time when green room becomes true. The outer F (Finally) operator ensures that this entire sequence occurs eventually during the system's execution. The conjunction (third_floor & F(green_room)) indicates that when the agent reaches the third floor, it is required that a subsequent visit to the green room will follow. In the natural language description, "go through the third floor and navigate to the green room" corresponds directly to this temporal ordering, where "go through the third floor" maps to third_floor and "navigate to the green room" maps to F(green_room), preserving the intended sequence of actions.
3. The formula ! (purple_room) U (third_floor) asserts that the condition third floor must eventually become true, and until that moment occurs, the condition purple room must remain false. The U (Until) operator specifies a temporal dependency in which the left-hand condition, ! (purple_room), must continuously hold up to the point when the right-hand condition, third_floor, becomes true. The negation ! (purple_room) enforces that the purple room is avoided before reaching the third floor. Once third floor is reached, the requirement of avoiding the purple room no longer applies,

meaning the agent may either remain on the third floor or subsequently visit the purple room. This directly corresponds to the natural language description, where “go to the third floor avoiding the purple room” maps to ! (purple_room) U (third_floor), and the allowance to “stay there, or head to the purple room” reflects the release of the avoidance constraint after the goal state is achieved.

4. The formula ! (green_room) U (landmark_2) specifies that the agent must eventually reach landmark 2, and up to that point, the condition green room must remain false. The U (Until) operator enforces a strict temporal ordering, requiring that the negation ! (green_room) holds continuously until the right-hand condition, landmark_2, becomes true. This means that entering the green room is prohibited before reaching landmark two. Once landmark 2 is reached, the restriction on entering the green room no longer applies. The natural language instruction “don’t go into the green room until going to landmark two” directly corresponds to this logic, with “don’t go into the green room” represented by ! (green_room) and “until going to landmark two” represented by U (landmark_2).
5. The formula F (third_floor & !green_room) states that at some point in the future, there will be a state where the agent is on the third floor while not being in the green room. The outer F (Finally) operator ensures that this combined condition being on the third floor and avoiding the green room must eventually occur during execution. The conjunction (third_floor & !green_room) requires that both conditions hold simultaneously at that future time. The negation !green_room specifies that the agent must not be in the green room at the moment the third floor is reached. Thus, while the natural language description says “always avoid the green room and navigate to the third floor,” the formula more precisely captures the requirement that the agent eventually reaches the third floor without being in the green room at that time, rather than continuously avoiding it throughout the entire trajectory.
6. The formula F (landmark_1 & !red_room) specifies that at some point in the future, the agent will reach a state where landmark 1 is true and red room is false. The outer F (Finally) operator ensures that this combination being at landmark one while not being in the red room must eventually occur. The conjunction (landmark_1 & !red_room) indicates that both conditions must

hold simultaneously at that future moment. The negation !red_room enforces that, at the time of reaching landmark one, the agent is not in the red room. Although the natural language instruction “avoid the red room and go to landmark one” suggests a continuous avoidance of the red room, the formula strictly guarantees only that the red room is not entered at the time the agent reaches landmark one.

7. The formula F (landmark_3) & G (third_floor) specifies that the agent must eventually reach landmark 3, while also ensuring that third floor remains true at all times. The conjunction (&) combines these two temporal requirements, indicating that both must hold within the same execution trace. The F (Finally) operator asserts that there will be some future point where landmark_3 becomes true, whereas the G (Globally) operator enforces that third_floor holds continuously throughout the entire timeline. This means the agent must always stay on the third floor while eventually reaching landmark three. The natural language command “go to landmark three but remain on the third floor” corresponds directly to this logical structure, mapping “go to landmark three” to F(landmark_3) and “remain on the third floor” to G(third_floor).
8. The formula F (landmark_1) & G (green_room) specifies that the agent must eventually reach landmark 1, while ensuring that green room remains true at all times during the process. The conjunction (&) connects these two temporal requirements, meaning both must be satisfied within the same execution. The F (Finally) operator guarantees that there exists a future moment when landmark_1 becomes true, and the G (Globally) operator enforces that the condition green_room holds continuously throughout the entire trajectory. This implies that the agent must stay within the green room at all times while eventually reaching landmark one. The natural language description “go to landmark one by only navigating within the green room” maps precisely to these components, with “go to landmark one” corresponding to F(landmark_1) and “only navigating within the green room” corresponding to G(green_room).
9. The formula F (green_room) specifies that at some point in the future, the condition green room will become true. The F (Finally) operator indicates that this event being in the green room must eventually occur, though not necessarily immediately. This means the agent is required to reach a state

where it enters or occupies the green room at least once during its execution. The natural language instruction “go to the green room” corresponds directly to this meaning, with “go to” representing the eventuality expressed by F and “the green room” mapping to the atomic proposition green_room.

10. The formula F (landmark_1) asserts that at some point in the future, the condition landmark 1 will become true. The F (Finally) operator specifies that the agent must eventually reach landmark one, without imposing any requirements on the states or actions taken before that occurs. This ensures that the system’s execution includes at least one moment in which the agent is at landmark one. The natural language command “navigate directly to landmark one” corresponds to the eventuality captured by F(landmark_1), where “navigate to” expresses the future achievement of reaching landmark_1.
11. The formula F (green_room & F (blue_room)) expresses that at some point in the future, the condition green room will become true, and from that moment onward, there will be a later time when blue_room also becomes true. The outer F (Finally) operator ensures that this entire sequence, reaching the green room and then eventually reaching the blue room must occur at least once in the system’s execution. The conjunction (green_room & F(blue_room)) indicates that when the agent enters the green room, it is required that a subsequent state satisfying blue_room will follow. This captures a temporal ordering in which the visit to the green room precedes the visit to the blue room. The natural language command “enter the blue room through the green room” directly maps to this logic, where “through the green room” corresponds to green_room occurring before F(blue_room), and “enter the blue room” corresponds to the eventual satisfaction of blue_room.
12. The formula F (green_room & F (blue_room)) expresses that at some point in the future, the condition green room will become true, and from that moment onward, there will be a later time when blue_room also becomes true. The outer F (Finally) operator ensures that this entire sequence, reaching the green room and then eventually reaching the blue room must occur at least once in the system’s execution. The conjunction (green_room & F(blue_room)) indicates that when the agent enters the green room, it is required that a subsequent state satisfying blue_room will follow. This

captures a temporal ordering in which the visit to the green room precedes the visit to the blue room. The natural language command “enter the blue room through the green room” directly maps to this logic, where “through the green room” corresponds to green_room occurring before F(blue_room), and “enter the blue room” corresponds to the eventual satisfaction of blue_room.

13. The formula F ((red_room ∨ yellow_room) & F (green_room)) specifies that at some point in the future, either red room or yellow room will become true, and from that moment onward, there will exist a later time when green room becomes true. The outer F (Finally) operator ensures that this entire sequence passing through one of the two rooms and then eventually reaching the green room must occur at least once during the execution. The disjunction (red_room ∨ yellow_room) indicates that visiting either the red room or the yellow room satisfies the initial part of the requirement, while the inner F (green_room) enforces that reaching the green room must follow. This temporal structure encodes a dependency where the agent first goes through one of the specified rooms before continuing to the green room. The natural language instruction “go through the yellow or red room to the green room” maps directly to this logic, with “yellow or red room” corresponding to (red_room ∨ yellow_room) and “to the green room” corresponding to F(green_room).
14. The formula F (blue_room) & G (!green_room) specifies that the agent must eventually reach the blue_room, while ensuring that green_room is never entered at any point in time. The F (Finally) operator guarantees that there exists a future moment when the condition blue_room becomes true, meaning the agent will at some point go to the blue room. The G (Globally) operator enforces that the negation !green_room holds throughout the entire execution, thereby prohibiting any visit to the green room. The conjunction (&) combines these two temporal requirements, ensuring that both the eventual arrival at the blue room and the continuous avoidance of the green room occur simultaneously. This directly corresponds to the natural language description “go to the blue room without going to the green room,” where “go to the blue room” maps to F(blue_room) and “without going to the green room” maps to G(!green_room).

15. The formula F (red_room) & G (!yellow_room) specifies that the agent must eventually reach the red room, while ensuring that the yellow room is never entered at any point during execution. The F (Finally) operator guarantees that a future state will occur in which red_room becomes true, representing the eventual visit to the red room. The G (Globally) operator enforces that the negation !yellow_room holds at all times, meaning the agent must continuously avoid the yellow room throughout the entire process. The conjunction (&) combines these temporal requirements, ensuring that both the eventual visit to the red room and the permanent avoidance of the yellow room are satisfied simultaneously. This directly reflects the natural language command "go to the red room but never go to the yellow room," where "go to the red room" corresponds to F(red_room) and "never go to yellow room" corresponds to G(!yellow_room).
16. The formula F (green_room) & G (!blue_room) specifies that the agent must eventually reach the green room, while continuously avoiding the blue room throughout the entire execution. The F (Finally) operator ensures that there will be a future moment when the condition green_room becomes true, indicating that the agent must at some point go to the green room. The G (Globally) operator enforces that the negation !blue_room holds at all times, prohibiting any entry into the blue room during the entire trajectory. The conjunction (&) combines these temporal requirements, ensuring both eventual arrival at the green room and persistent avoidance of the blue room. This precisely captures the natural language directive "go to the green room but avoid the blue room," where "go to the green room" maps to F(green_room) and "avoid the blue room" maps to G(!blue_room).
17. The formula F (red_room) specifies that at some point in the future, the condition red room will become true. The F (Finally) operator denotes eventuality, meaning that the agent is required to reach a state where it enters or occupies the red room at least once during its execution. This ensures that the red room is visited eventually, though the formula imposes no constraints on when this must occur or what happens before or after. The natural language instruction "go to the red room" corresponds directly to this meaning, with "go to" representing the temporal eventuality expressed by F and "the red room" corresponding to the atomic proposition red_room.

18. The formula F (blue_room) states that at some point in the future, the condition blue room will become true. The F (Finally) operator expresses eventuality, requiring that the agent must eventually reach a state where it enters or is located in the blue room. This means that the blue room must be visited at least once during the system's execution, though no constraints are placed on the timing or the states preceding or following this event. The natural language description "enter the blue room" directly corresponds to this meaning, where "enter" represents the eventual satisfaction implied by F, and "the blue room" maps to the atomic proposition blue_room.
19. The formula F ((red_room ∨ yellow_room) & F (blue_room)) specifies that at some point in the future, the agent will reach a state where either red room or yellow room becomes true, and from that point onward, there will exist a later state in which blue room becomes true. The outer F (Finally) operator ensures that this entire sequence passing through one of the two rooms and then eventually entering the blue room occurs at least once in the execution. The disjunction (red_room ∨ yellow_room) indicates that visiting either the red room or the yellow room satisfies the first part of the condition, while the inner F (blue_room) enforces that reaching the blue room must follow subsequently. The natural language command "go through the yellow or red room to reach the blue room" maps precisely to this structure, where "yellow or red room" corresponds to (red_room ∨ yellow_room) and "reach the blue room" corresponds to F(blue_room).
20. The formula F ((red_room ∨ blue_room) & F (green_room)) specifies that at some point in the future, the agent will reach a state where either red room or blue room becomes true, and from that state onward, there will exist a later moment when green room becomes true. The outer F (Finally) operator ensures that this entire sequence visiting one of the two rooms followed by eventually reaching the green room must occur at least once during execution. The disjunction (red_room ∨ blue_room) allows either the red room or the blue room to fulfill the first part of the condition, while the inner F (green_room) enforces that the green room must be reached afterward. The natural language command "go through the red or blue room to the green room" corresponds directly to this logical structure, where "red or blue room" maps to (red_room ∨ blue_room) and "to the green room" maps to F(green_room).

21. The formula F (Citizen's Bank & F (Chipotle)) specifies that at some point in the future, the condition Citizen's Bank will become true, and from that moment onward, there will exist a later time when Chipotle becomes true. The outer F (Finally) operator ensures that this entire sequence first reaching Citizen's Bank and then eventually reaching Chipotle occurs at least once during the system's execution. The conjunction (Citizen's Bank & F(Chipotle)) requires that when the agent reaches Citizen's Bank, there must still be a future moment in which Chipotle is subsequently found. This establishes a clear temporal ordering between the two locations. The natural language instruction "find Citizen's Bank and then find Chipotle" directly corresponds to this structure, with "find Citizen's Bank" represented by Citizen's Bank and "then find Chipotle" represented by F(Chipotle).
22. The formula F (CVS & F (Watson Center)) expresses that at some point in the future, the condition CVS will become true, and from that state onward, there will be a later time when Watson Center becomes true. The outer F (Finally) operator guarantees that this entire sequence first visiting CVS and then eventually reaching Watson Center must occur at least once in the system's execution. The conjunction (CVS & F(Watson Center)) ensures that when the agent arrives at CVS, there remains a future state in which Watson Center is subsequently visited. The natural language instruction "go to CVS and then go to Watson Center" corresponds directly to this logical formulation, where "go to CVS" maps to CVS and "then go to Watson Center" maps to F(Watson Center).
23. The formula F (Barus Building & F (CVS)) specifies that at some point in the future, the condition Barus Building will become true, and from that point onward, there will exist a later moment when CVS becomes true. The outer F (Finally) operator ensures that this entire sequence first reaching the Barus Building and then eventually reaching CVS must occur at least once during the execution. The conjunction (Barus Building & F(CVS)) indicates that when the agent reaches the Barus Building, there must still be a future state where CVS is subsequently found. This establishes a strict temporal order, requiring the Barus Building to be encountered before CVS. The natural language instruction "find Barus Building and then find CVS" corresponds directly to this logic, with "find Barus Building" represented by Barus Building and "then find CVS" represented by F(CVS).

24. The formula G (!Angell St) & F (CVS) specifies that the agent must never enter Angell St while ensuring that CVS is eventually reached. The G (Globally) operator enforces that the negation !Angell St holds at all times, meaning the condition of not being on Angell Street must remain true throughout the entire execution. The F (Finally) operator requires that at some point in the future, the condition CVS becomes true, indicating that the agent must eventually reach CVS. The conjunction (&) combines these two temporal requirements, ensuring continuous avoidance of Angell Street while also guaranteeing the eventual arrival at CVS. This directly aligns with the natural language instruction "do not enter Angell St and find CVS," where "do not enter Angell St" maps to G(!Angell St) and "find CVS" maps to F(CVS).
25. The formula G (!Angell St) & F (Barus Building) specifies that the agent must always avoid Angell St while eventually reaching the Barus Building. The G (Globally) operator enforces that the negation !Angell St remains true at all times, meaning the agent is prohibited from entering Angell Street throughout the entire execution. The F (Finally) operator ensures that there exists a future point where Barus Building becomes true, representing the requirement to eventually find or reach that location. The conjunction (&) combines these two temporal conditions, ensuring continuous avoidance of Angell Street alongside the eventual visit to the Barus Building. The natural language instruction "stay away from Angell St and find Barus Building" directly corresponds to this logic, with "stay away from Angell St" represented by G(!Angell St) and "find Barus Building" represented by F(Barus Building).
26. The formula G (!Brook St) & F (Bookstore & F (Science Library)) specifies that the agent must continuously avoid Brook St while ensuring that it first reaches the Bookstore and subsequently the Science Library. The G (Globally) operator enforces that the negation !Brook St holds at all times, meaning the agent must never enter Brook Street during execution. The F (Bookstore & F (Science Library)) component requires that, at some future point, the Bookstore is reached, and from that state onward, there will be a later moment when the Science Library is reached, establishing a temporal order between the two locations. The natural language instruction "avoid Brook St and first go to Bookstore and then Science Library" directly

corresponds to this formulation, where “avoid Brook St” maps to G(!Brook St), “first go to Bookstore” maps to Bookstore, and “then Science Library” maps to F(Science Library).

27. The formula G (Nelson Fitness Center) & F (Chipotle & F (Nelson Fitness Center)) specifies that the condition Nelson Fitness Center must hold continuously throughout the entire execution while also ensuring a sequential relationship in which Chipotle is reached, followed by another occurrence of Nelson Fitness Center. The G (Globally) operator enforces that Nelson Fitness Center remains true at all times, indicating that the agent must always be in or maintain access to that location. The F (Chipotle & F (Nelson Fitness Center)) component ensures that there will be a future point where Chipotle is reached, and from that point onward, there will later be another state where Nelson Fitness Center is again visited. The conjunction (&) requires both the continuous truth of Nelson Fitness Center and the eventual sequence involving Chipotle. This directly reflects the natural language instruction “always visit Nelson Fitness Center, and find Chipotle after that,” where “always visit Nelson Fitness Center” corresponds to G(Nelson Fitness Center) and “find Chipotle after that” corresponds to F(Chipotle & F(Nelson Fitness Center)).
28. The formula G (Science Library) & F (Starbucks & F (Science Library)) specifies that the agent must always remain within the Science Library while also ensuring that at some point, Starbucks is visited and later the Science Library is again encountered. The G (Globally) operator enforces that the condition Science Library holds continuously throughout the execution, indicating that the agent is persistently located there. The F (Starbucks & F (Science Library)) component requires that eventually a state is reached where Starbucks becomes true, and from that point onward, there will be a later state where Science Library is again true, establishing a return after visiting Starbucks. The natural language instruction “stay in Science Library and then go to Starbucks” corresponds to this structure, with “stay in Science Library” represented by G(Science Library) and “then go to Starbucks” represented by F(Starbucks & F(Science Library)).
29. The formula G (Barus Building) & F (Watson Center & F (Barus Building)) specifies that the agent must continuously remain in the Barus Building while ensuring that at some point, it visits the Watson Center and later

returns to the Barus Building. The G (Globally) operator enforces that Barus Building holds at all times, meaning the agent must persistently stay within or maintain presence in that location throughout execution. The F (Watson Center & F (Barus Building)) component requires that eventually a state will occur where Watson Center is reached, and from that point onward, there will later be another state where Barus Building is again true, expressing a temporal dependency between visiting Watson Center and returning. The conjunction (&) combines these conditions, requiring both the continuous truth of Barus Building and the eventual sequence involving Watson Center. The natural language instruction "remain in Barus Building and then visit Watson Center" maps directly to this logic, where "remain in Barus Building" corresponds to G(Barus Building) and "visit Watson Center" corresponds to F(Watson Center & F(Barus Building)).

30. The formula G (!Brook St) & F (Science Library) specifies that the agent must never enter Brook St while ensuring that it eventually reaches the Science Library. The G (Globally) operator enforces that the negation !Brook St remains true at all times, meaning the agent must continuously avoid Brook Street throughout the entire execution. The F (Science Library) component guarantees that at some point in the future, the condition Science Library will become true, indicating that the agent must eventually find or arrive at the Science Library. The conjunction (&) combines these two temporal requirements, ensuring both persistent avoidance of Brook Street and the eventual achievement of reaching the Science Library. The natural language instruction "do not enter Brook St but find Science Library" corresponds directly to this logic, where "do not enter Brook St" maps to G(!Brook St) and "find Science Library" maps to F(Science Library).
31. The Linear Temporal Logic formula G (¬Angell St) ∧ F (FedEx Office) specifies that throughout the entire timeline, the condition "Angell St" must never be true, and at some point in the future, the condition "FedEx Office" must eventually become true. The operator G (globally) enforces that the negation of "Angell St" holds at all times, corresponding to the natural language instruction to "avoid Angell St." The operator F (finally) indicates that "FedEx Office" will be satisfied at least once in the future, aligning with the directive to "find FedEx Office." The conjunction ∧ combines these two temporal requirements, ensuring that avoidance of Angell St is maintained

continuously while the eventual discovery of FedEx Office is guaranteed. Thus, the formula precisely captures a persistent prohibition of one condition and the eventual fulfillment of another.

32. The Linear Temporal Logic formula F (Bookstore) asserts that the proposition “Bookstore” will become true at some point in the future. The temporal operator F (finally) specifies that there exists at least one future moment when the condition “Bookstore” holds. This directly corresponds to the natural language instruction “Go to Bookstore,” which implies that reaching the Bookstore must eventually occur, though not necessarily immediately. In essence, it captures the guarantee of eventually arriving at the Bookstore.
33. The Linear Temporal Logic formula F (Chipotle) expresses that the proposition “Chipotle” will eventually become true at some point in the future. The temporal operator F (finally) signifies that there exists a future state where the condition “Chipotle” holds. This corresponds to the natural language statement “Find where is Chipotle,” indicating that the goal is to eventually reach or identify the location of Chipotle. The formula imposes no requirement on when this occurs, only that it must happen at least once in the future. Thus, it formally guarantees the eventual realization of the state associated with Chipotle.
34. The Linear Temporal Logic formula G (Science Library) $\wedge$ F (CVS) specifies two distinct temporal requirements: a persistent condition and an eventual goal. The operator G (globally) requires that the proposition “Science Library” holds at all times, corresponding to the instruction to “always stay in Science Library.” The operator F (finally) asserts that “CVS” will become true at least once in the future, matching the phrase “eventually go to CVS.” The conjunction $\wedge$ combines these two conditions, enforcing that the state of being in the Science Library is continuously maintained while ensuring that a future transition to CVS ultimately occurs. Therefore, the formula expresses the simultaneous persistence of one condition and the eventual satisfaction of another.
35. The formula G (Watson Center) & F (Starbucks) specifies that the agent must continuously remain in the Watson Center while also ensuring that it will eventually reach Starbucks at some point in the future. The G (Globally) operator enforces that the condition Watson Center holds at all times,

meaning the agent must stay within or maintain presence at Watson Center throughout the execution. The F (Finally) operator requires that a future state will occur where Starbucks becomes true, indicating that the agent must eventually go to or reach Starbucks. The conjunction (&) combines these two temporal constraints, requiring both continuous presence in Watson Center and the eventual satisfaction of the condition Starbucks. The natural language instruction "remain at Watson Center but eventually go to Starbucks" maps directly to this logic, where "remain at Watson Center" corresponds to G(Watson Center) and "eventually go to Starbucks" corresponds to F(Starbucks).

36. The formula G((iwBoSK & NpPUaHQZqPQa) → G(!IM_CcZj)) specifies that whenever both iwBoSK and NpPUaHQZqPQa become true at the same time, it must thereafter always hold that IM_CcZj is false. The outer G (Globally) operator ensures that this rule applies at all times, meaning the implication must be valid throughout the entire execution. The inner implication (iwBoSK & NpPUaHQZqPQa) → G(!IM_CcZj) establishes a conditional dependency: once both antecedent conditions are satisfied, the consequent G(!IM_CcZj) enforces a permanent prohibition on IM_CcZj for all future states. The nested G(!IM_CcZj) expresses an unending restriction, ensuring that the forbidden condition never becomes true once the triggering event occurs. This precisely corresponds to the natural language statement "after both iwBoSK and NpPUaHQZqPQa, absolutely never IM_CcZj," where the co-occurrence of the two initial conditions activates a permanent global ban on IM_CcZj.
37. The formula G(!PXjEPOci) requires that the proposition PXjEPOci is false at every point along the execution trace, i.e., PXjEPOci must never occur. The G (Globally) operator enforces that the enclosed property holds for all times, and the negation !PXjEPOci specifies the property to be the non-occurrence of PXjEPOci. The natural language statement "under no condition PXjEPOci" corresponds directly to G(!PXjEPOci), with "under no condition" mapping to the global negation that forbids PXjEPOci at all times.
38. The formula G(!(ckojym_vNUR | mOqug)) | F(ckojym_vNUR & F(mOqug)) asserts that at least one of two alternatives holds: either both ckojym_vNUR and mOqug are permanently absent, or there is a temporal

sequence in which ckojym_vNUR occurs and is later followed by mOqug. The left disjunct G(!(ckojym_vNUR | mOqug)) uses G (Globally) together with a negated disjunction to require that neither proposition ever becomes true at any time. The right disjunct F(ckojym_vNUR & F(mOqug)) uses an outer F (Finally) to require that at some future time ckojym_vNUR holds, and the embedded F(mOqug) enforces that after that time there exists a later moment when mOqug holds. The top-level disjunction (|) combines these alternatives, so the trace is acceptable if it either always avoids both propositions or eventually realizes ckojym_vNUR followed subsequently by mOqug; the natural language phrase "eventually, ckojym_vNUR or mOqug and, at some point later, mOqug again" corresponds to the second alternative F(ckojym_vNUR & F(mOqug)), while the first alternative captures the possibility of never observing either proposition.

39. The formula F(qeBtdFQHmb) asserts that at some point in the future, the condition qeBtdFQHmb will become true. The F (Finally) operator expresses eventuality, meaning that the system's execution must include at least one state where qeBtdFQHmb holds. This temporal operator does not specify when the event occurs, only that it must occur at some future time. The natural language statement "it will happen that qeBtdFQHmb" directly corresponds to this meaning, where "it will happen" reflects the eventual occurrence guaranteed by F, and "qeBtdFQHmb" denotes the proposition that must eventually hold.
40. The formula G((aWpSUYgiTCy) → G((CAcfkLQXAL) → F(dvWRlxPo))) specifies that if aWpSUYgiTCy ever becomes true, then from that point onward, it must always hold that whenever CAcfkLQXAL occurs, dvWRlxPo will eventually follow. The outer G (Globally) operator ensures that this rule is valid at all times throughout the system's execution. The inner implication (CAcfkLQXAL → F(dvWRlxPo)) establishes a conditional relationship in which the occurrence of CAcfkLQXAL necessitates the eventual satisfaction of dvWRlxPo, as enforced by the F (Finally) operator. The natural language statement "aWpSUYgiTCy involves that whenever CAcfkLQXAL occurs, dvWRlxPo eventually happens" corresponds precisely to this logic, where "involves that whenever" reflects the conditional structure enforced by the nested G operators and "eventually happens" maps to F(dvWRlxPo).

41. The formula G((sYCGfBAVVp ∨ QkktvCMxcVGI) → F(QXjvGmzx)) specifies that at all times, if either sYCGfBAVVp or QkktvCMxcVGI becomes true, then QXjvGmzx must eventually occur afterward. The outer G (Globally) operator ensures that this conditional requirement applies throughout the entire execution. The disjunction (sYCGfBAVVp ∨ QkktvCMxcVGI) states that the triggering event can be either of the two propositions, and the implication requires that whenever one of them occurs, it activates the obligation expressed on the right-hand side. The operator F (Finally) in F(QXjvGmzx) demands that a future state must make QXjvGmzx true, thereby establishing a guaranteed eventual consequence. The natural language description "as either sYCGfBAVVp or QkktvCMxcVGI, finally, QXjvGmzx will happen" directly corresponds to this structure, with the disjunction mapping to the triggering condition and "finally" mapping to the eventuality enforced by F(QXjvGmzx).
42. The formula G(vUnUHzC) specifies that the proposition vUnUHzC must hold at every point throughout the system's execution. The G (Globally) operator enforces that the condition inside it remains true at all times, without exception. This directly matches the natural language description "it is always true that vUnUHzC," where "always true" corresponds precisely to the global constraint expressed by G.
43. The formula G((qYWhgQe) → G(mIEYuyhBx & vWAYect)) specifies that at all times, whenever qYWhgQe becomes true, it must thereafter always be the case that both mIEYuyhBx and vWAYect hold. The outer G (Globally) operator ensures that this implication is continuously enforced throughout the entire execution. The implication (qYWhgQe → …) establishes that the truth of qYWhgQe serves as a trigger for the requirement on the right-hand side. The inner G(mIEYuyhBx & vWAYect) then requires that from the moment qYWhgQe occurs onward, the conjunction of mIEYuyhBx and vWAYect must remain true at every subsequent state. The natural language description "whenever qYWhgQe then every time both mIEYuyhBx and vWAYect hold" maps directly to this structure, where "whenever" corresponds to the implication and "every time" corresponds to the inner global operator.
44. The formula G((GgvGADnCuwzUs) → F(IgbAcsPKo | cenfclGT)) specifies that at all times, whenever GgvGADnCuwzUs becomes true, there

must later be a state in which either IgbAcsPKo or cenfclGT holds. The outer G (Globally) operator ensures that this conditional requirement is enforced throughout the entire execution. The implication establishes that the occurrence of GgvGADnCuwzUs triggers an obligation on the right-hand side, and the disjunction (IgbAcsPKo | cenfclGT) indicates that satisfying either proposition is sufficient to meet that obligation. The natural language description "every time GgvGADnCuwzUs then in the future either IgbAcsPKo or cenfclGT" corresponds directly to this structure, with "every time" mapping to the global implication and "in the future" mapping to the eventuality expressed by F.

45. The formula G((paPQJQPqQOPwgn) → F(GXjrgYRkt_Gn_ | FEsX)) specifies that at all times, if paPQJQPqQOPwgn becomes true, then at some later point either GXjrgYRkt_Gn_ or FEsX must occur. The outer G (Globally) operator ensures that this implication is enforced continuously throughout the entire execution. The implication establishes that the truth of paPQJQPqQOPwgn triggers the requirement on the right-hand side, and the disjunction (GXjrgYRkt_Gn_ | FEsX) indicates that satisfying either proposition is sufficient to meet the obligation. The natural language description "whenever paPQJQPqQOPwgn then sooner or later GXjrgYRkt_Gn_ or FEsX" directly reflects this structure, with "whenever" corresponding to the implication and "sooner or later" corresponding to the eventuality expressed by F.
46. The formula G(!(sTqTp | zEmMXnN | ideleiA_fgg)) specifies that none of the propositions sTqTp, zEmMXnN, or ideleiA_fgg may ever become true at any point in the execution. The G (Globally) operator requires the enclosed condition to hold at all times, ensuring a permanent restriction. The negation applies to the entire disjunction, meaning that each of the three propositions must remain false continuously. The natural language statement "it will not happen that sTqTp or zEmMXnN or ideleiA_fgg" corresponds directly to this global constraint, with the phrase "will not happen" mapping to G(! …) and the listed events mapping to the disjunction inside the negation.
47. The formula G(tjVXmfFhIx | AQogUdFHEZkp | mdPPpTdDtZrebD) states that at every point in the execution, at least one of the three propositions tjVXmfFhIx, AQogUdFHEZkp, or mdPPpTdDtZrebD must be true. The G

(Globally) operator enforces that this requirement holds continuously, without exception, across all future states. The disjunction ensures that any one of the listed propositions is sufficient to satisfy the condition at each moment, but at least one must always be active. The natural language statement “eternally, either tjVXmfFhIx, AQogUdFHEZkp or mdPPpTdDtZrebD” corresponds directly to this meaning, with “eternally” mapping to G and the listed alternatives mapping to the disjunction inside it.

48. The formula F(AjZXYF_zM | KfDeQSGCPTEw | BPZqUyExRRq) specifies that at some point in the future, at least one of the propositions AjZXYF_zM, KfDeQSGCPTEw, or BPZqUyExRRq must become true. The F (Finally) operator requires that the disjunction it governs be satisfied in some future state, though it imposes no constraints on when this occurs. The disjunction itself indicates that any one of these three propositions is sufficient to fulfill the requirement. The natural language description “finally, either AjZXYF_zM, KfDeQSGCPTEw or BPZqUyExRRq” maps directly to this meaning, with “finally” corresponding to F and the listed propositions corresponding to the disjunction.
49. The formula G((oGeoKTIhZKz_W) → F(lOGrS & BgSIpcmj)) specifies that at all times, whenever oGeoKTIhZKz_W becomes true, there must later be a state in which both lOGrS and BgSIpcmj hold simultaneously. The outer G (Globally) operator ensures that this implication is enforced throughout the entire execution. The implication itself states that the occurrence of oGeoKTIhZKz_W triggers the obligation expressed on the right-hand side. The operator F (Finally) requires that a future state satisfy the conjunction lOGrS & BgSIpcmj, meaning both propositions must eventually be true together after the triggering condition. The natural language description “if oGeoKTIhZKz_W then it will happen that both lOGrS and BgSIpcmj” directly maps to this structure, with “if” corresponding to the implication and “it will happen” corresponding to the eventuality expressed by F.
50. The formula G((dbQVDo & uotjbfVtn) → G(!syBUlnQ)) specifies that at all times, if dbQVDo and uotjbfVtn are both true simultaneously, then from that point onward syBUlnQ must remain false. The outer G (Globally) operator ensures that this implication applies continuously throughout the entire execution. The conjunction (dbQVDo & uotjbfVtn) serves as the

triggering condition, and once it occurs, the inner G(!syBUlnQ) requires that syBUlnQ is never allowed to occur in any future state. The natural language description “always both dbQVDo and uotjbfVtn implies that it will not happen that syBUlnQ” maps directly to this structure, with the implication reflecting the conditional relationship and “will not happen” expressed by the global negation G(!syBUlnQ).

51. The formula G(!JLOlbbwc) | F(JLOlbbwc & F(TkfnNgDBvtbM)) states that either JLOlbbwc never occurs at any time, or there is a future moment when JLOlbbwc becomes true and, from that moment onward, a later state occurs in which TkfnNgDBvtbM becomes true. The left disjunct G(!JLOlbbwc), governed by the G (Globally) operator, enforces the permanent absence of JLOlbbwc if this branch of the disjunction is satisfied. The right disjunct F(JLOlbbwc & F(TkfnNgDBvtbM)) uses the outer F (Finally) operator to require a future state where JLOlbbwc holds, and the nested F(TkfnNgDBvtbM) ensures that TkfnNgDBvtbM must then occur at some later point. The natural language description “first, in the future JLOlbbwc, and then, in the future TkfnNgDBvtbM” corresponds to this second branch, which establishes the ordered eventualities encoded by the two nested F operators.
52. The formula G((zFVJLgTpxjAEHN) → G((Z_AuRh) → F(nKIlX))) specifies that whenever zFVJLgTpxjAEHN holds, it activates a persistent obligation that thereafter whenever Z_AuRh occurs, nKIlX must eventually follow. The outer G (Globally) operator enforces that this conditional rule applies at all times along the execution trace, and the outer implication (zFVJLgTpxjAEHN → …) makes the obligation contingent on the truth of zFVJLgTpxjAEHN. The nested G((Z_AuRh) → F(nKIlX)) requires that from the moment zFVJLgTpxjAEHN is true, the implication (Z_AuRh → F(nKIlX)) must hold at every subsequent state; within that implication, F(nKIlX) (Finally) demands that after any occurrence of Z_AuRh there exists a later state in which nKIlX becomes true. The natural language statement “zFVJLgTpxjAEHN involves that if Z_AuRh then it will happen that nKIlX” maps directly to this structure, with “zFVJLgTpxjAEHN involves” corresponding to the antecedent that enables the rule (zFVJLgTpxjAEHN → …), “if Z_AuRh” corresponding to the conditional

(Z_AuRh → …), and “it will happen that nKIlX” corresponding to the eventuality F(nKIlX).

53. The formula F(cfZvISle) specifies that at some point in the future, the proposition cfZvISle must become true. The F (Finally) operator expresses eventuality, requiring that the system’s execution contains at least one state in which cfZvISle holds. The natural language statement “eventually, cfZvISle” corresponds directly to this meaning, with “eventually” mapping to the F operator and “cfZvISle” mapping to the atomic proposition that must occur.
54. The formula G(ERIzBflUWqY | ETbuydMOmBLV | JVygMUXkXT) specifies that at every point in the execution, at least one of the propositions ERIzBflUWqY, ETbuydMOmBLV, or JVygMUXkXT must hold. The G (Globally) operator requires that this condition be true at all times, without exception. The disjunction inside the operator means that any one of the three propositions is sufficient to satisfy the requirement at each state, but the system may not enter a state where all three are false. No additional temporal structure is involved, so the formula imposes a single continuous constraint across the entire timeline. The natural language statement “every time either ERIzBflUWqY, ETbuydMOmBLV or JVygMUXkXT” corresponds directly to this meaning, with “every time” mapping to the global requirement expressed by G, and the listed options matching the disjunction inside it.
55. The formula G(!(ApZGIwL | dXwhbCPbJVMuwUb | vAvKToY)) specifies that none of the propositions ApZGIwL, dXwhbCPbJVMuwUb, or vAvKToY may ever be true at any point in the execution. The G (Globally) operator enforces that this restriction holds continuously, ensuring that the negated disjunction remains true at all times. The negation applies to the entire disjunction, meaning that every state must avoid all three propositions simultaneously. The natural language statement “it is never the case that ApZGIwL or dXwhbCPbJVMuwUb or vAvKToY” corresponds directly to this global constraint, with “never” mapping to G(! …) and the listed propositions mapping to the disjunction inside the negation.
56. The formula G(!(escalator_moves & house_is_open & engine_starts)) specifies that at no point in the execution may all three conditions: escalator_moves, house_is_open, and engine_starts be true simultaneously.

The G (Globally) operator requires this prohibition to hold continuously across all states, ensuring that the forbidden combination is never allowed to occur. The negation applies to the entire conjunction, meaning that the system must avoid any state in which all three propositions are true at the same time. The natural language statement "it is never the case that the escalator moves and together the house is open and the engine starts" corresponds directly to this meaning, with "never the case" mapping to G(! …) and the listed conditions mapping to the conjunction inside the negation.

57. The formula G(car_enters & train_derails & car_stops) specifies that at every point in the system's execution, the three conditions car_enters, train_derails, and car_stops must all hold simultaneously. The G (Globally) operator requires this conjunction to remain true at all times, imposing a continuous constraint on the system's behavior. The conjunction car_enters & train_derails & car_stops means that each of the three propositions must be true in every state, with no exceptions. The natural language description "all the time a car enters and together a train derails and the car stops" maps directly to this structure, with "all the time" corresponding to G and the three listed events corresponding to the conjunctive condition inside it.
58. The formula F(table_has_been_moved & train_has_been_launched & motorbike_falls_down) specifies that at some point in the future there will be a state in which all three conditions: table has been moved, train has been launched, and motorbike falls down hold simultaneously. The F (Finally) operator requires that this combined event occurs at least once, though it does not impose any constraint on when it must happen. The conjunction inside the operator ensures that the three propositions must be true together in the same state, not merely at different times. The natural language description "it is going to happen that a table has been moved and, at the same time, the train has been launched and a motorbike falls down" corresponds directly to this meaning, with "it is going to happen" mapping to F and the listed events mapping to the conjunction.
59. The formula G((car_starts & engine_starts) → F(constructor_instantiate_objects & engine_stops)) specifies that at all times, if a state occurs in which both car starts and engine starts hold simultaneously, then there must later be a state in which constructor instantiate objects and engine stops are both true together. The outer G

(Globally) operator ensures that this implication is enforced continuously throughout the execution. The conjunction car_starts & engine_starts forms the triggering condition, and once it becomes true, the F (Finally) operator on the right-hand side requires that a future state satisfy the conjunction constructor_instantiate_objects & engine_stops. The natural language description "whenever both the car starts and the engine starts then at a certain moment the constructor instantiate objects and the engine stops" corresponds directly to this structure, with "whenever" mapping to the implication and "at a certain moment" mapping to the eventuality expressed by F.

60. The formula G((brake_is_released | table_is_brown) → G(!(constructor_instantiate_objects & house_is_open))) specifies that at all times, if either brake is released or table is brown becomes true, then from that moment onward it must always be the case that the conjunction constructor instantiate objects & house is open is false. The outer G (Globally) operator ensures that this conditional rule applies continuously throughout the entire execution. The disjunction brake_is_released | table_is_brown serves as the trigger, and once one of these conditions holds, the inner G(!(...)) requires a permanent prohibition on any state where both constructor_instantiate_objects and house_is_open are simultaneously true. The natural language description "whenever either the brake is released or a table is brown then never, a constructor instantiate objects and a house is open" maps directly to this structure, with "whenever" corresponding to the implication and "never" corresponding to the inner global negation.
61. The Linear Temporal Logic formula G (¬(table_is_brown | motorbike_has_started | motorbike_has_stopped)) expresses that at all times, none of the three specified conditions are ever true. The operator G (globally) enforces that this restriction holds continuously throughout the entire timeline. The negation ¬ applies to the disjunction, ensuring that the propositions "table_is_brown," "motorbike_has_started," and "motorbike_has_stopped" are each false at every moment. The logical operator ∨ (or) groups these propositions, representing the natural language phrase "either the table is brown, a motorbike has started, or a motorbike has stopped." Consequently, the formula captures the requirement that it must always be the case that none of these events occur under any circumstance.

62. The Linear Temporal Logic formula G (escalator_speeds_up & brake_is_pressed & engine_starts) states that all three listed propositions must hold at every moment in time. The operator G (globally) enforces this persistent requirement, corresponding directly to the natural language phrase "always." The conjunction ∧ links the propositions "escalator_speeds_up," "brake_is_pressed," and "engine_starts," reflecting the phrasing "at the same time" in the description. Because the formula places no temporal modifiers other than G, it requires these three conditions to be simultaneously true at every state along the timeline. Thus, the formula expresses a continuous and unbroken concurrence of the escalator speeding up, the brake being pressed, and the engine starting.
63. The Linear Temporal Logic formula F (motorbike_falls_down & motorbike_catches_fire & engine_starts) states that there will be some moment in the future at which all three listed propositions hold simultaneously. The operator F (finally) ensures the eventual occurrence of such a state, aligning with the natural language phrase "it is going to happen." The conjunction ∧ links "motorbike_falls_down," "motorbike_catches_fire," and "engine_starts," reflecting the requirement that these events occur at the same time. The formula imposes no constraints on the current or intermediate states, only guaranteeing that one future state satisfies all three conditions together. Thus, it formally captures the eventual simultaneous realization of the motorbike falling down, catching fire, and having its engine start.
64. The Linear Temporal Logic formula G((bar_is_down ∧ brake_is_released) → F(train_derails ∧ brake_is_released)) states that at every moment, if the conditions "bar_is_down" and "brake_is_released" hold simultaneously, then there must eventually be a future moment in which both "train_derails" and "brake_is_released" are true. The operator G (globally) ensures that this implication is evaluated at all times, corresponding to the natural language phrase "whenever." The conjunction ∧ in the antecedent captures the simultaneous occurrence of the bar being down and the brake being released, while the implication → expresses that this situation triggers a required future outcome. The operator F (finally) then guarantees that there will be a later state in which the train derails while the brake remains released, matching the phrase "at some point a train derails and the brake is released."

Thus, the formula specifies a persistent rule linking a present joint condition with a required future joint outcome.

65. The Linear Temporal Logic formula G((sensor_captures_data | brake_is_released) → G(brake_is_released | motorbike_has_started)) states that at every moment, if either "sensor_captures_data" or "brake_is_released" is true, then a further requirement must hold continuously thereafter. The outer operator G (globally) enforces that this implication is evaluated at all times, corresponding to the natural language condition "if … then in any case." The disjunction ∨ in the antecedent reflects the phrase "the sensor captures data or the brake is released," meaning that either proposition is sufficient to trigger the implication. The inner operator G ensures that once the antecedent is satisfied, the disjunction "brake_is_released ∨ motorbike_has_started" must hold at all subsequent states, aligning with the statement that "the brake is released or the motorbike has started" in every case. Thus, the formula specifies a persistent rule in which a present disjunctive condition entails a continuous future disjunctive requirement.
66. The Linear Temporal Logic formula G(¬(brake_is_pressed | motorbike_has_started)) | F((brake_is_pressed | motorbike_has_started) ∧ F(table_is_brown | motorbike_has_started)) expresses a disjunction between a persistent prohibition and a structured eventual sequence of events. The left disjunct uses G (globally) together with negation to require that neither "brake_is_pressed" nor "motorbike_has_started" ever occurs, representing the possibility that these events do not happen at any time. The right disjunct employs F (finally) to assert that there is a future moment when either the brake is pressed or a motorbike has started, matching the natural language phrase "first, at some point in time the brake is pressed or a motorbike has started." Within this future moment, the conjunction ∧ links this initial event to another F operator, which ensures that at a later time it will happen that either a table is brown or the motorbike has started, aligning with the phrase "and then, it will happen that a table is brown or the motorbike has started."
67. The Linear Temporal Logic formula G((semaphore_is_green) → G((sensor_retrieves_data) → F(house_is_built))) states that at every moment, if the proposition "semaphore_is_green" holds then a further,

persistent requirement is imposed thereafter. The outer operator G (globally) ensures the implication is evaluated at all times, while the antecedent (semaphore_is_green) names the triggering condition; the implication → requires that whenever that trigger is true the consequent must hold. The inner G enforces that from the moment the semaphore is green the conditional (sensor_retrieves_data → F(house_is_built)) must hold continuously, where F (finally) requires that whenever sensor_retrieves_data is true at some time, there exists a later time at which house_is_built becomes true. The natural language phrase "when a semaphore is green" maps to the antecedent of the outer implication, "always when the sensor retrieves data" corresponds to the inner G and the antecedent sensor_retrieves_data, and "at some point the house is built afterwards" corresponds to the inner F(house_is_built); no additional temporal or causal assumptions are implied beyond these operators.

68. The Linear Temporal Logic formula G((semaphore_is_broken ∧ train_is_crossing) → F(bridge_opens ∧ train_has_been_launched)) specifies that at every point in time, if the semaphore is broken and the train is crossing simultaneously, then a particular future condition must eventually occur. The operator G (globally) requires that this implication is evaluated at all moments, corresponding to the natural language phrase "whenever." The conjunction ∧ in the antecedent captures the simultaneous truth of "semaphore_is_broken" and "train_is_crossing," reflecting the condition described in the natural language statement. The operator F (finally) in the consequent ensures that there will be some future moment in which both "bridge_opens" and "train_has_been_launched" hold together, matching the phrase "it will happen that the bridge opens and a train has been launched." Thus, the formula expresses a persistent rule linking a present joint condition to an eventual joint outcome.
69. The Linear Temporal Logic formula G(sensor_gathers_information | house_is_built) asserts that at every moment in time, at least one of the two propositions must hold. The operator G (globally) enforces that this requirement applies continuously, corresponding to the natural language phrase "in any case." The disjunction ∨ captures the condition that either "sensor_gathers_information" or "house_is_built" must be true, directly reflecting the wording "either the sensor gathers information or a house is

built." Because no additional temporal operators appear inside the formula, it imposes a simple, ongoing constraint rather than a sequential or eventual one. Thus, the formula expresses an uninterrupted guarantee that one of the two specified propositions is always satisfied.

70. The Linear Temporal Logic formula G(¬(car_enters | car_stops | engine_starts)) specifies that none of the listed propositions are ever permitted to occur at any point in time. The operator G (globally) requires this condition to hold continuously, corresponding to the natural language directive "never." The negation ¬ applies to the entire disjunction, ensuring that "car_enters," "car_stops," and "engine_starts" are all false at every moment. The disjunction ∨ groups these propositions together, directly reflecting the phrase "the car enters or the car stops or the engine starts." Thus, the formula expresses a strict, uninterrupted prohibition against any of these events ever becoming true.
71. The formula G(!(cpu_cools_down & monitor_lights_up & model_is_underfitted)) specifies that at no point during the system's execution may all three conditions: cpu cools down, monitor lights up, and model is underfitted be true simultaneously. The G (Globally) operator requires that this prohibition hold continuously, ensuring that every state in the execution avoids this specific combination. The negation applies to the entire conjunction, meaning that while each proposition may occur individually, they may never occur together in the same state. The natural language description "it should not happen that a cpu cools down and together the monitor lights up and a model is underfitted" corresponds directly to this meaning, with "should not happen" mapping to G(! …) and the listed events mapping to the conjunctive condition inside the negation.
72. The formula G(model_overfits & computer_stops & monitor_lights_up) specifies that at every point in the execution, the three conditions model_overfits, computer_stops, and monitor_lights_up must all hold simultaneously. The G (Globally) operator enforces that this requirement is true at all times, without exception. The conjunction inside the operator requires that each of the three propositions be true together in every state, not merely at different moments. The natural language description "it is always the case that the model overfits and, at the same time, a computer stops and the monitor lights up" maps directly to this meaning, with "always

the case" corresponding to G and the three listed events corresponding to the conjunctive condition.

73. The formula F(monitor_lights_up & model_overfits & fridge_is_switched_on) specifies that at some point in the future, there must be a state in which monitor lights up, model overfits, and fridge is switched on all simultaneously. The F (Finally) operator requires the eventual occurrence of such a state but does not constrain when it must happen. The conjunction inside the operator ensures that all three propositions must be true together in the same moment, rather than at different times. The natural language description "finally, the monitor lights up and together a model overfits and the fridge is switched on" corresponds directly to this meaning, with "finally" mapping to F and the listed events mapping to the conjunctive expression.
74. The formula G((model_underfits) → F(plane_lands | computer_starts)) specifies that at all times, whenever model underfits becomes true, there must later be a state in which either plane lands or computer starts holds. The G (Globally) operator ensures that this conditional requirement applies continuously throughout the entire execution. The implication states that the occurrence of model_underfits triggers the obligation on the right-hand side, and the disjunction plane_lands ∨ computer_starts indicates that either proposition is sufficient to satisfy that obligation. The natural language description "after a model underfits, sooner or later either the plane lands or the computer starts" corresponds directly to this structure, with "after" mapping to the implication and "sooner or later" mapping to the eventuality expressed by F.
75. The formula G(!(computer_starts & model_is_underfitted)) | F((computer_starts & model_is_underfitted) & F(model_is_underfitted & cpu_is_cooled_down)) states that either the combination computer starts & model is underfitted never occurs, or there is a future moment in which this combination does occur and is later followed by another state where model is underfitted and cpu is cooled down hold simultaneously. The left disjunct G(!(computer_starts & model_is_underfitted)) uses the G (Globally) operator together with a negated conjunction to enforce the permanent absence of the initial event combination. The right disjunct begins with F((computer_starts & model_is_underfitted) & …), where the outer F

(Finally) requires that the system eventually reaches a state where both computer_starts and model_is_underfitted are true together. Inside that state, the nested F(model_is_underfitted & cpu_is_cooled_down) requires an additional later state in which model_is_underfitted and cpu_is_cooled_down again hold simultaneously. The natural language description "finally, a computer starts and a model is underfitted and, at a certain moment both a model is underfitted and a cpu is cooled down afterwards" corresponds to the second branch, which imposes this ordered pair of eventualities.

76. The formula G((cpu_overheats) → G((mouse_captures_input) → F(cpu_is_overheated))) specifies that at all times, if cpu overheats becomes true, it activates a persistent requirement governing later states. The outer G (Globally) operator ensures that this rule applies continuously throughout the execution. Once cpu_overheats holds, the inner G((mouse_captures_input) → …) requires that at every subsequent state, if mouse_captures_input occurs, it must be followed by a future state in which cpu_is_overheated becomes true. The operator F (Finally) in F(cpu_is_overheated) expresses this eventual requirement, ensuring that the overheating condition must occur after each triggering mouse-input event. The natural language description "when the cpu overheats then whenever a mouse captures input then at some point in time the cpu is overheated afterwards" corresponds directly to this structure, with "when" mapping to the outer implication, "whenever" to the inner implication, and "at some point in time" to the eventuality expressed by F.
77. The formula G((computer_stops | model_underfits) → F(cpu_overheats & mouse_captures_input)) specifies that at all times, if either computer stops or model underfits becomes true, then there must later be a state in which cpu overheats and mouse captures input hold simultaneously. The outer G (Globally) operator ensures that this implication applies throughout the entire execution, so the requirement is continuously enforced. The disjunction computer_stops | model_underfits serves as the trigger, and the implication states that whenever either condition occurs, the system becomes obligated to satisfy the right-hand side. The operator F (Finally) requires that a future state exist where both cpu_overheats and mouse_captures_input are true together, as enforced by the conjunction inside it. The natural language

statement “if the computer stops or a model underfits then eventually, both a cpu overheats and the mouse captures input” corresponds directly to this structure, with “if” mapping to the implication, “or” mapping to the disjunction, and “eventually” mapping to the F operator.

78. The formula G(computer_stops & monitor_shows_output & fridge_is_switched_on) specifies that at every point in the system’s execution, the three conditions computer stops, monitor shows output, and fridge is switched on must all hold simultaneously. The G (Globally) operator requires that this combined condition be true at all times, without exception. The conjunction inside the operator enforces that each of the three propositions must occur together in every state, not merely at different times. The natural language description “at any time a computer stops and together the monitor shows output and a fridge is switched on” corresponds directly to this meaning, with “at any time” mapping to G and the listed events mapping to the conjunctive expression.
79. The formula G((computer_stops | mouse_clicks) → G(!(plane_lands & fridge_is_switched_on))) specifies that at all times, if either computer stops or mouse clicks becomes true, then from that moment onward it must always be the case that plane lands and fridge is switched on are not true simultaneously. The outer G (Globally) operator enforces that this implication is checked continuously throughout the entire execution. The disjunction computer_stops ∨ mouse_clicks serves as the triggering condition, and once it holds, the inner G(!(plane_lands & fridge_is_switched_on))) requires a permanent prohibition of the conjunction of plane_lands and fridge_is_switched_on. The natural language description “always either a computer stops or the mouse clicks implies that it will not happen that a plane lands and the fridge is switched on” corresponds directly to this structure, with “always” mapping to the outer G, the triggering phrase mapping to the disjunction, and “will not happen” mapping to the inner global negation.
80. The formula G(!(computer_starts | fridge_is_empty | monitor_shows_output)) specifies that none of the propositions computer starts, fridge is empty, or monitor shows output may ever be true at any point in the execution. The G (Globally) operator requires the enclosed condition to hold at all times, establishing a continuous and unbroken restriction. The

negation applies to the entire disjunction, meaning that every state must avoid all three events simultaneously; any one of them becoming true would violate the formula. The natural language description “under no circumstances a computer starts or the fridge is empty or the monitor shows output” maps directly to this meaning, with “under no circumstances” corresponding to G(! …) and the listed events matching the disjunction inside the negation.